

§7.7 - Approximate Integration

Problem

We want to approximate

$$\int_a^b f(x) \, dx.$$

Definition #1 (Uniform Partition)

Let $[a, b] \subset \mathbf{R}$, $n \in \mathbf{N}$ and

$$\Delta x = \frac{b - a}{n}.$$

Then

$$P = \{x_0, x_1, \dots, x_n\}$$

where

$$x_k = a + k\Delta x = x_{k-1} + \Delta x$$

is a **uniform partition** of $[a, b]$ with **step-size** Δx .

Definition #2 (Left Riemann Sum)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} L_n &= \sum_{k=1}^n f(x_{k-1})\Delta x \\ &= (f(x_0) + f(x_1) + \dots + f(x_{n-1}))\Delta x \end{aligned}$$

is a **left Riemann sum** approximation of $\int_a^b f$.

Definition #3 (Right Riemann Sum)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} R_n &= \sum_{k=1}^n f(x_k) \Delta x \\ &= (f(x_1) + \dots + f(x_n)) \Delta x \end{aligned}$$

is a **right Riemann sum** approximation of $\int_a^b f$.

Remark

We have

$$L_n - R_n = (f(x_0) - f(x_n))\Delta x = (f(a) - f(b))\Delta x$$

and

$$\int_a^b f(x) \, dx \approx L_n \approx R_n.$$

Definition #4 (Midpoint Riemann Sum)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$M_n = \sum_{k=1}^n f(\bar{x}_k) \Delta x = (f(\bar{x}_1) + \dots + f(\bar{x}_n)) \Delta x$$

where

$$\bar{x}_k = x_{k-1} + \Delta x/2 = \frac{x_{k-1} + x_k}{2}$$

is a **midpoint Riemann sum** approximation of $\int_a^b f$.

Definition #5 (Trapezoidal Rule)

Assume $[a, b]$ is an interval, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$\begin{aligned} T_n &= \left(\frac{f(x_0)}{2} + \sum_{k=1}^{n-1} f(x_k) + \frac{f(x_n)}{2} \right) \Delta x \\ &= (f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)) \frac{\Delta x}{2} \end{aligned}$$

is a **trapezoidal rule** approximation of $\int_a^b f$.

Remark

Naturally,

$$\int_a^b f(x) \, dx \approx T_n$$

and

$$T_n = \frac{L_n + R_n}{2},$$

but T_n is **not** a Riemann sum.

Example #1

Compute

$$T_2, \quad M_2, \quad \text{and} \quad T_4$$

for

$$\int_0^2 e^x \, dx.$$

Definition #6 (Error Terms)

Assume M_n and T_n are the midpoint and trapezoidal rule approximations of $\int_a^b f$.

Then

$$E_M(n) = \int_a^b f(x) \, dx - M_n \quad \text{and} \quad E_T(n) = \int_a^b f(x) \, dx - T_n$$

are the **error terms** of the approximations.

Theorem #1

Let f'' be continuous on $[a, b]$ and assume M_n and T_n are midpoint and trapezoidal rule approximations of $\int_a^b f$.

Then

$$|E_M(n)| \leq \frac{K(b-a)^3}{24n^2} = \frac{K(b-a)}{24}(\Delta x)^2$$

and

$$|E_T(n)| \leq \frac{K(b-a)^3}{12n^2} = \frac{K(b-a)}{12}(\Delta x)^2$$

where $|f''(x)| \leq K$ for all $x \in [a, b]$.

Definition #7 (Simpson's Rule)

Assume $[a, b]$ is an interval, n is even, $\Delta x = (b - a)/n$, and f is integrable on $[a, b]$.

Then

$$S_n = (f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \dots \\ + 4f(x_{n-3}) + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)) \frac{\Delta x}{3}$$

is a **Simpson's rule** approximation of $\int_a^b f$.

Theorem #2

Let $f^{(4)}$ be continuous on $[a, b]$ and assume S_n is a Simpson's rule approximation of $\int_a^b f$.

Then

$$E_S(n) = \int_a^b f(x) \, dx - S_n$$

satisfies

$$|E_S(n)| \leq \frac{K(b-a)^5}{180n^4} = \frac{K(b-a)}{180}(\Delta x)^4.$$

where $|f^{(4)}(x)| \leq K$ for all $x \in [a, b]$.

Example #2

Estimate

$$E_T(10) \quad \text{and} \quad E_S(6)$$

for

$$\int_0^2 e^x \, dx.$$

Example #3

Estimate

$$E_S(4) \quad \text{and} \quad E_S(6)$$

for

$$\int_0^4 x^3 \, dx.$$

Example #4

Find

$$|E_T(8)| \quad \text{and} \quad |E_S(8)|$$

for

$$\int_0^2 e^x \, dx.$$

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}
128	1×10^{-4}	2×10^{-9}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}
128	1×10^{-4}	2×10^{-9}
256	3×10^{-5}	1×10^{-10}

n	$ E_T(n) $	$ E_S(n) $
2	5×10^{-1}	3×10^{-2}
4	1×10^{-1}	2×10^{-3}
8	3×10^{-2}	1×10^{-4}
16	8×10^{-3}	9×10^{-6}
32	2×10^{-3}	5×10^{-7}
64	5×10^{-4}	3×10^{-8}
128	1×10^{-4}	2×10^{-9}
256	3×10^{-5}	1×10^{-10}
512	8×10^{-6}	8×10^{-12}